Intelli-Credit — Requirements Specification

Version 2.0 | Unified Specification (Original + Updated PS)

Section 1: Product Overview

What Intelli-Credit Is

Intelli-Credit is an AI-powered Credit Decisioning Engine that automates end-to-end preparation of a Credit Appraisal Memo (CAM) for Indian corporate and SME lending. It is delivered as a **hosted web application** (accessible without VPN) that transforms raw, unstructured financial data into a comprehensive, AI-backed investment/assessment report.

The system ingests multiple types of data — structured financial files, unstructured documents, live web research, and human observations — runs them through a multi-layer intelligence pipeline with **human-in-the-loop validation** at critical junctions, and produces a professional, fully explainable lending recommendation with SWOT analysis.

The system behaves like a digital senior credit manager. Not a black box that spits out a score, but a reasoning system that can be interrogated at every step, where every conclusion traces back to evidence, and where humans approve, deny, or edit AI classifications before final analysis.

What It Is Not

- **Not a Loan Origination System.** It plugs into existing LOS infrastructure as a decisioning brain.
- **Not a fully automated approve/reject machine.** Humans remain in control of the final decision and validate AI-driven data extraction.
- **Not a generic credit scoring tool.** Every element is built around Indian lending context, Indian data sources, and Indian regulatory nuance.

Core Design Principles

Explainability first. Every number, signal, and recommendation must be traceable back to a source document, a rule, or a model output. No black boxes anywhere.

Human-in-the-loop validation. AI classifies and extracts, but humans approve, deny, or edit critical classifications and schema mappings before analysis proceeds.

Policy-aligned. Different lenders have different risk appetites. The system is configurable at three distinct levels.

Indian context built-in. GST reconciliation logic, MCA data structures, e-Courts parsing, CIBIL Commercial nuances, ALM analysis, portfolio performance tracking — first-class citizens, not add-ons.

360-degree view. Secondary research goes beyond news to include legal disputes, market sentiment, sector trends, macro indicators, and any alternate data that provides material insight.

Section 2: System Architecture Overview

The system has **four sequential layers** plus a **pre-processing onboarding layer**. Each layer's output feeds the next, and every handoff is visible and auditable.

LAYER 0 — ENTITY ONBOARDING
(Capture entity details + loan parameters)
↓
Raw Documents (5 core types + supporting docs) + External Sources + Primary Notes
↓
LAYER 1 — DATA INGESTOR
(Classify, Parse, Extract, Schema Map with Human Validation)
↓
LAYER 2 — ML INTELLIGENCE LAYER
(FinBERT + Isolation Forest + Graph Engine)
↓
LAYER 3 — FIVE Cs SCORECARD + PRE-COGNITIVE ANALYSIS
(Aggregate signals → scores → decision + SWOT)
↓
LAYER 4 — LLM SYNTHESIS
(CAM narrative + SWOT + recommendation)
↓
Interactive UI + Word/PDF CAM with SWOT Matrix

Every layer's output is stored, labeled, and linked forward and backward. A judge can start at the final CAM recommendation and drill all the way back to the exact sentence in a PDF that triggered a risk signal. **That is the explainability promise.**

Section 0: Entity Onboarding (NEW)

0.1 Purpose

Before any document upload or analysis, the system must capture structured entity metadata and loan parameters. This ensures proper classification, sector-specific model selection (Isolation Forest), and context-aware analysis.

0.2 Onboarding Flow

Multi-step form interface (not a single page — better UX):

Step 1: Entity Identification

Field	Type	Validation	Purpose
Company Name	Text	Required	Display throughout system
CIN (Corporate Identification Number)	Text (21 chars)	Format: L/U + 5 digits + State code + Year + PLC/PTC + 6 digits	Unique identifier, MCA lookup
PAN	Text (10 chars)	Format: 5 letters + 4 digits + 1 letter	Tax identifier, ITR cross-check
Sector	Dropdown	Manufacturing / NBFC / Retail / Infrastructure / Pharma / Other	Determines Isolation Forest model
Sub-sector	Text	Optional	Granular sector analysis
State of Registration	Dropdown	Indian states	Jurisdiction, e-Courts filtering
Annual Turnover	Number (INR Crores)	>0	Size classification, ratio benchmarks

Step 2: Loan Details

Field	Type	Validation	Purpose
Loan Type	Dropdown	Term Loan / Working Capital / Cash Credit / Letter of Credit / Bank Guarantee	Determines covenant structure
Proposed Amount	Number (INR)	>0	DSCR back-calculation, limit suggestion
Tenure	Number (months)	>0	Amortization schedule

Proposed Interest Rate	Number (% per annum)	Optional	Risk premium comparison
Purpose of Loan	Text	Required	Context for capacity assessment

Step 3: Document Upload (Expanded from original drawer)

See Section 3 for full document types. UI allows batch upload with drag-and-drop.

Step 4: Processing Initiation

"Analyze" button triggers full pipeline. Processing overlay shows sequential steps (retained from original design).

0.3 Data Storage

All onboarding data stored in:

Table: `case_metadata`

Column	Type	Notes
case_id	STRING (PK)	Auto-generated unique ID
company_name	STRING	Display name
cin	STRING	Validated format
pan	STRING	Validated format
sector	ENUM	Determines ML model
sub_sector	STRING	Optional
state	STRING	For e-Courts filtering
annual_turnover_cr	DECIMAL(10,2)	Size indicator
loan_type	ENUM	Facility type
proposed_amount_inr	DECIMAL(15,2)	Loan quantum
tenure_months	INTEGER	Loan tenure
proposed_interest_rate_pct	DECIMAL(5,2)	If provided
loan_purpose	TEXT	Business justification
onboarded_by	STRING	User who created case

onboarding_timestamp	TIMESTAMP	Case creation time

Section 3: Inputs — Full Coverage

These are every input type the system must handle, mapped to their source and what they feed into.

3.1 Structured Inputs

(Unchanged from original specification)

- **GST Filings (GSTR-2A and GSTR-3B)**
- **ITRs (Income Tax Returns)**
- **Bank Statements**

Full details in comprehensive data specification (previous output).

3.2 Unstructured Inputs

UPDATED: 5 Core Document Types (New PS) + Original Supporting Documents

CORE DOCUMENTS (Priority Upload)

1. ALM (Asset-Liability Management) Statement

What it is: A financial report showing the maturity profile of assets (loans, investments) vs liabilities (deposits, borrowings). Critical for NBFCs and banks to assess liquidity risk and interest rate risk.

Format: PDF or Excel, often multi-tab spreadsheet

What we extract:

- Maturity buckets (1-7 days, 8-14 days, 15-30 days, 1-3 months, 3-6 months, 6-12 months, 1-3 years, >3 years)
- Assets by bucket: cash, liquid investments, loan receivables maturing
- Liabilities by bucket: deposits maturing, borrowings due
- Cumulative gaps (assets - liabilities) by bucket
- Negative gap flags (liabilities > assets in critical short-term buckets)

Feeds into:

- **Capacity C** (liquidity risk signal — if short-term liabilities >> short-term assets, cash crunch risk)
- **Capital C** (asset quality assessment from maturity structure)

- **Isolation Forest** (ALM gap ratios as features)

Table: `alm_data`

Column	Type	Notes
case_id	STRING	Case identifier
maturity_bucket	ENUM	1-7d / 8-14d / 15-30d / 1-3m / 3-6m / 6-12m / 1-3y / >3y
assets_bucket_inr	DECIMAL(15,2)	Assets maturing in bucket
liabilities_bucket_inr	DECIMAL(15,2)	Liabilities due in bucket
cumulative_gap_inr	DECIMAL(15,2)	Running gap (assets - liabilities)
gap_as_pct_of_assets	DECIMAL(5,2)	Gap / Total Assets × 100
negative_gap_flag	BOOLEAN	true if liabilities > assets
extraction_confidence	DECIMAL(5,2)	0-100 from Textract
source_document_id	STRING (FK)	Source ALM file

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2. Shareholding Pattern

What it is: Mandatory disclosure (for listed companies) showing ownership structure. **Critically, shows promoter pledge %** (shares pledged as collateral for personal loans).

Format: PDF or Excel from BSE/NSE filings

What we extract:

- Promoter holding %
- Promoter pledge % (**CRITICAL** — >60% is high risk)
- Institutional holding trends (FII, DII, Mutual Funds)
- Public holding %
- Significant ownership changes (>2% movement quarter-over-quarter)

Feeds into:

- **Character C** — promoter financial stress signal (pledge % is top-weighted signal)
- **Capital C** — ownership stability

Table: `shareholding_pattern_quarterly` (as defined in comprehensive spec)

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3. Borrowing Profile

What it is: Comprehensive list of all existing debt facilities — term loans, working capital limits, non-fund based (LC/BG), from all lenders. Shows debt structure, covenants, and payment track record.

Format: PDF (often internal credit bureau report or sanction letter compilation) or Excel

What we extract:

- Lender name, facility type, sanctioned amount, outstanding amount
- Interest rate (high rates = risk signal)
- Security/collateral taken
- Covenants imposed
- Any restructuring history, waivers, defaults
- Payment track record (regular / irregular / arrears)

Feeds into:

- **Capital C** — total debt calculation (add off-balance-sheet facilities)
- **Collateral C** — existing encumbrances (reduce available collateral)
- **Character C** — if restructuring/defaults mentioned

Table: `borrowing_profile`

Column	Type	Notes
case_id	STRING	Case identifier
lender_name	STRING	Bank/NBFC/Institution
facility_type	ENUM	term_loan / working_capital / cash_credit / letter_of_credit / bank_guarantee / overdraft
sanctioned_amount_inr	DECIMAL(15,2)	Approved limit
outstanding_amount_inr	DECIMAL(15,2)	Current dues
interest_rate_pct	DECIMAL(5,2)	**High rates = risk**
security_description	TEXT	Collateral taken
covenants	TEXT	Financial/operational covenants
restructuring_flag	BOOLEAN	Ever restructured
payment_track_record	ENUM	regular / irregular / arrears / default
source_document_id	STRING (FK)	Source borrowing profile file

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4. Annual Reports (P&L, Cash Flow, Balance Sheet)

What it is: Audited financials (3 years) containing Profit & Loss, Cash Flow Statement, Balance Sheet, plus auditor notes, management discussion, risk factors, corporate governance.

Format: PDF, 100-300 pages

What we extract:

- **Financial Statements (3 years):** Revenue, EBITDA, PAT, Total Assets, Total Liabilities, Net Worth, Cash Flow from Operations
- **Ratios computed:** DSCR, Debt/Equity, Interest Coverage, Current Ratio, EBITDA Margin, Revenue CAGR, PAT Margin
- **Auditor qualifications** (going concern flags, emphasis of matter)
- **Related party transactions**
- **Contingent liabilities**
- **Management commentary** on risks

Feeds into:

- **All Five Cs**
- **Isolation Forest** (all 7 ratios)
- **FinBERT** (auditor notes, risk disclosures)

Tables: `financial_statements_line_items`, `computed_financial_ratios`, `auditor_notes`, `related_party_transactions`, `contingent_liabilities` (as defined in comprehensive spec)

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5. Portfolio Cuts / Performance Data

What it is: For NBFCs/lenders, this shows the performance of their loan portfolio — segment-wise NPA, restructured assets, vintage analysis, concentration risk.

Format: PDF or Excel with portfolio analytics

What we extract:

- **Portfolio composition:** Secured vs unsecured, retail vs corporate, sector-wise split
- **Asset quality metrics:** Gross NPA %, Net NPA %, restructured asset %
- **Vintage analysis:** Loans originated by year, their performance
- **Concentration:** Top 10 borrowers as % of portfolio, top 5 sectors
- **Provisioning coverage ratio**

Feeds into:

- **Capacity C** — asset quality indicates earning sustainability
- **Capital C** — provisioning requirements reduce net worth
- **Conditions C** — portfolio concentration = sector exposure risk

Table: `portfolio_performance`

Column	Type	Notes
case_id	STRING	Case identifier
portfolio_category	STRING	Secured / Unsecured / Retail / Corporate / Sector
portfolio_outstanding_inr	DECIMAL(15,2)	Total book
gross_npa_pct	DECIMAL(5,2)	% of portfolio in NPA
net_npa_pct	DECIMAL(5,2)	After provisioning
restructured_assets_pct	DECIMAL(5,2)	% restructured
top_10_borrower_concentration_pct	DECIMAL(5,2)	Concentration risk
provisioning_coverage_ratio	DECIMAL(5,2)	Provisions / Gross NPAs
source_document_id	STRING (FK)	Source portfolio file

SUPPORTING DOCUMENTS (Original PS)

- **Board Meeting Minutes**
- **Rating Agency Reports**
- **Sanction Letters from Other Banks**

Full details in comprehensive data specification.

3.3 External Intelligence

EXPANDED: Beyond News to Alternate Data

News Reports (Original)

- **Source:** Tavily API
- **Coverage:** Company, promoters, sector trends, regulatory changes
- **Feeds into:** Character C, Conditions C

MCA Filings (Ministry of Corporate Affairs) — EXPANDED

- **What it is:** Official company registry containing directors, charges (loans), company status
- **NEW ADDITION:** Fetch full MCA profile including:
    - Annual filing compliance status
    - Disqualified directors check
    - Related party network (common directors across companies)
- **Feeds into:** Character C, Graph Engine

e-Courts Legal Disputes — EXPANDED

- **What it is:** Active and disposed court cases
- **NEW ADDITION:** Broader search including:
    - Civil cases (not just DRT)
    - Cheque bounce cases under Negotiable Instruments Act
    - Consumer disputes
    - Arbitration proceedings
- **Feeds into:** Character C (litigation density)

Market Sentiment Analysis (NEW)

- **Source:** Social media scraping (Twitter/X, LinkedIn), industry forums, analyst reports
- **What we extract:** Sentiment polarity (positive/negative/neutral), trending discussions about company/sector
- **Feeds into:** Conditions C (market perception risk)

Macro Economic Indicators (NEW)

- **Source:** RBI website, Government portals
- **What we extract:**
    - Sector-specific growth rates
    - Regulatory changes (RBI circulars, SEBI notifications)
    - Interest rate trends
    - Inflation data
- **Feeds into:** Conditions C (macro headwinds/tailwinds)

3.4 Primary Insights

(Unchanged from original specification)

- **Factory Site Visit Observations** (structured dropdowns)
- **Management Interview Notes** (structured form mapped to Five Cs)

Section 4: Layer 1 — Data Ingestor

4.1 Purpose

Layer 1 converts all raw inputs into clean, structured signals. **NEW ADDITION:** Human-in-the-loop validation ensures AI classifications are approved before analysis proceeds.

4.2 Document Classification (NEW)

Process:

1. **Auto-Classification:** Upon upload, ML model (fine-tuned document classifier or rule-based on filename/content) assigns document type:
    - ALM Statement
    - Shareholding Pattern
    - Borrowing Profile
    - Annual Report
    - Portfolio Cuts
    - GST Filing
    - Bank Statement
    - ITR
    - Board Minutes
    - Rating Report
    - Sanction Letter
    - Other/Unknown
2. **Human Validation (CRITICAL):**
    - UI displays classification in **Ingestor page** with confidence score
    - User sees: "Classified as: ALM Statement (89% confidence)"
    - User options:
        - **Approve** (proceed with this classification)
        - **Deny** (mark as misclassified, select correct type from dropdown)
        - **Edit** (change to different type)
    - **No analysis proceeds until all documents are approved**

Table: `document_classification`

Column	Type	Notes
document_id	STRING (PK)	Unique document
case_id	STRING	Case identifier
original_filename	STRING	As uploaded
auto_classified_type	ENUM	AI's classification

classification_confidence	DECIMAL(5,2)	0-100
human_validated_type	ENUM	After user approval/edit
validation_status	ENUM	pending / approved / denied / edited
validated_by	STRING	User who validated
validation_timestamp	TIMESTAMP	When validated
denial_reason	TEXT	If denied, why

4.3 OCR Pipeline

(Unchanged from original — AWS Textract with confidence scoring <75% flagged)

4.4 Dynamic Schema Configuration (NEW)

Purpose: Allow users to define or configure how extracted data is structured, especially for non-standard formats.

UI Flow:

1. After OCR completes, system shows **Schema Editor** in Ingestor page
2. Left panel: **Field Configuration**
    - Pre-defined schema templates for each document type (e.g., ALM has maturity buckets)
    - User can:
        - Add custom fields
        - Rename fields
        - Change data types (text / number / date / boolean)
        - Mark fields as required/optional
3. Right panel: **Extracted Data Table**
    - Shows actual data extracted from document
    - Cells are **editable** (user can correct OCR errors)
    - Color-coded by confidence: green (>85%), amber (75-85%), red (<75%)
4. **Mapping:** User drags extracted columns to schema fields (if auto-mapping failed)
5. **Validate & Save:** User approves schema, data ingested into database

Example: ALM Document

Pre-defined schema has fields: `maturity_bucket`, `assets_bucket_inr`, `liabilities_bucket_inr`

Extracted table has columns: "Bucket", "Assets (₹Cr)", "Liabilities (₹Cr)"

User maps:

- "Bucket" → `maturity_bucket`
- "Assets (₹Cr)" → `assets_bucket_inr` (system auto-converts Cr to absolute)
- "Liabilities (₹Cr)" → `liabilities_bucket_inr`

User edits cell in "Assets (₹Cr)" row 3 from "12.5" to "125" (OCR missed decimal place)

Table: `schema_mappings`

Column	Type	Notes
mapping_id	STRING (PK)	Unique mapping
case_id	STRING	Case identifier
document_type	ENUM	Which doc type
schema_template_version	STRING	Schema version used
field_mappings	JSON	{extracted_col: schema_field}
custom_fields_added	JSON	User-defined fields
manual_edits_applied	JSON	Cell-level corrections
validated_by	STRING	Who approved
validation_timestamp	TIMESTAMP	When approved

4.5 Table Extraction and Financial Spreading

(Unchanged from original — 7 ratios computed for 3 years)

4.6 GST vs Bank Reconciliation

(Unchanged from original — revenue reliability score, circular trading detection)

4.7 Document-Specific Extraction Rules

(Expanded to include 5 new document types)

ALM Statement:

- Extract maturity buckets (regex patterns: "1-7 days", "8-14 days", etc.)
- Parse asset/liability columns
- Calculate cumulative gaps
- Flag negative gaps in short-term buckets (<30 days)

Shareholding Pattern:

- Extract promoter pledge % (keyword: "pledged" + percentage in same row/paragraph)
- Flag if >40% (caution), >60% (high risk)

Borrowing Profile:

- Extract lender names, amounts, rates
- Flag restructuring keywords ("OTS", "waiver", "moratorium", "restructuring")
- High interest rate flag (>Prime + 500 bps)

Portfolio Cuts:

- Extract NPA percentages
- Parse concentration tables (Top 10 borrowers)
- Calculate provisioning coverage ratio

Original documents: Board Minutes, Rating Reports, Sanction Letters (rules unchanged)

4.8 Databricks Role

(Unchanged — orchestration, Delta tables, parallel processing, data lineage)

Section 5: Layer 2 — ML Intelligence Layer

(Unchanged from original specification)

Three components:

1. **FinBERT Document Risk Extractor** (zero-shot classification)
2. **Isolation Forest Anomaly Detector** (trained on Screener.in data, sector-specific models)
3. **Graph Risk Engine** (NetworkX + Cytoscape.js visualization)

Full details in comprehensive data specification.

Section 6: Layer 3 — Five Cs Scorecard + Pre-Cognitive Analysis

6.1 Purpose

The scorecard aggregates all signals from Layer 2, Layer 1 outputs, and primary notes into structured, explainable scores. **NEW ADDITION:** Pre-Cognitive Analysis layer identifies forward-looking risk signals beyond traditional credit metrics.

6.2 The Five Cs — What Each Measures

(Unchanged from original — Character, Capacity, Capital, Collateral, Conditions)

Signal additions from new document types:

Character C:

- Shareholding pattern: promoter pledge %
- Borrowing profile: restructuring history

Capacity C:

- ALM data: liquidity gap ratios
- Portfolio cuts: asset quality (NPA %)

Capital C:

- Borrowing profile: total debt (add off-balance sheet)
- Portfolio cuts: provisioning requirements

Collateral C:

- Borrowing profile: existing encumbrances

Conditions C:

- Market sentiment analysis
- Macro economic indicators

6.3 Pre-Cognitive Analysis (NEW)

What it is: Forward-looking risk signals that traditional credit analysis misses. Identifies early warning signals before they appear in financial statements.

Components:

1. **Velocity Trends:** Month-over-month changes in key metrics
    - Revenue velocity (accelerating/decelerating)
    - Debt velocity (increasing/stable/decreasing)
    - NPA velocity (for NBFCs)
2. **Triangulation Flags:** Cross-verification across data sources
    - GST vs ITR vs Financial Statements revenue comparison
    - Borrowing profile debt vs Balance Sheet debt
    - Promoter interview claims vs documentary evidence
3. **Sentiment Divergence:** Qualitative vs quantitative mismatch
    - Management interview tone (confident) vs financial performance (declining)
    - News sentiment (negative) vs rating agency outlook (stable)
4. **Network Contagion Risk:** Graph-derived propagation risk
    - If borrower's major customer has stress signals → borrower at risk
    - If director's other companies have DRT cases → promoter stress

Table: `precognitive_signals`

Column	Type	Notes
signal_id	STRING (PK)	Unique signal
case_id	STRING	Case identifier
signal_category	ENUM	velocity_trend / triangulation_flag / sentiment_divergence / network_contagion
signal_description	TEXT	Plain-English explanation
severity	ENUM	low / medium / high / critical
data_sources_used	JSON	Which inputs triggered signal
feeds_into_c	ENUM	Which C this affects
score_impact	DECIMAL(3,2)	-2.0 to +2.0

Example:

signal_category: "triangulation_flag"
signal_description: "Revenue reported in ITR (₹95Cr) is 12% lower than Financial Statement revenue (₹108Cr). Suggests potential ITR underreporting or financial statement inflation."
severity: "medium"
data_sources_used: ["ITR FY2023-24", "Annual Report FY2023-24"]
feeds_into_c: "Character"

score_impact: -0.8

6.4 Scoring Structure

(Unchanged from original — each C out of 10, weighted composite score)

6.5 Three-Level Configurability System

(Unchanged from original — Level 1: C-level weights, Level 2: Signal weights, Level 3: Per-case overrides)

6.6 Decision Output

(Unchanged — score bands: Approve / Approve with Conditions / Refer / Reject + limit suggestion + risk premium + "what would change this decision")

Section 7: Layer 4 — LLM Synthesis

7.1 What an LLM Is and Why We Use One Here

(Unchanged — Claude/GPT-4 for narrative generation, not decision-making)

7.2 What the LLM Receives as Input

EXPANDED JSON payload includes:

- Original inputs (unchanged)
- **NEW:** Entity onboarding metadata (CIN, PAN, sector, loan details)
- **NEW:** Pre-cognitive signals
- **NEW:** Schema validation status (were any classifications denied?)
- **NEW:** Alternate data findings (market sentiment, macro indicators)

7.3 What the LLM Produces

Expanded CAM Structure:

Section 1 — Executive Summary (unchanged)

Section 2 — Company and Promoter Profile

- **NEW:** Include CIN, PAN, entity onboarding details
- **NEW:** Shareholding structure (from shareholding pattern doc)

Section 3 — Industry and Conditions

- **NEW:** Macro economic indicators relevant to sector
- **NEW:** Market sentiment analysis (social media, forums)

Section 4 — Financial Analysis

- **NEW:** ALM gap analysis (if applicable for NBFCs)
- **NEW:** Portfolio performance analysis (if applicable)

Section 5 — Five Cs Narrative (unchanged)

Section 6 — Pre-Cognitive Risk Analysis (NEW)

- Velocity trends identified
- Triangulation flags (cross-source mismatches)
- Sentiment divergence signals
- Network contagion risks

Section 7 — SWOT Analysis (NEW)

AI-generated SWOT matrix:

Strengths:

- Strong points from Five Cs (e.g., "DSCR 1.8x, well above policy minimum")
- Positive graph signals (e.g., "No director-level litigation")
- Favorable sector trends

Weaknesses:

- Risk signals from Five Cs (e.g., "Promoter pledge 68%, high risk threshold")
- Negative FinBERT findings
- Asset quality issues (if portfolio cuts show high NPA)

Opportunities:

- Sector growth drivers (from macro data, news)
- Expansion plans mentioned in management interview
- Diversification potential

Threats:

- Regulatory headwinds (RBI circulars, sector stress)

- Competitive pressures (from news, sentiment analysis)
- Network contagion risks (from graph)

Format: 2×2 matrix with 3-5 bullet points per quadrant

User Editability: In UI, SWOT matrix is **editable** — user can add/remove/modify points before finalizing CAM

Section 8 — Proposed Facility and Structure (unchanged)

Section 9 — Recommendation and Rationale

- **NEW:** Include pre-cognitive signals in reasoning if material
- **NEW:** Reference SWOT when explaining decision

Section 10 — Audit Trail Summary (unchanged)

Section 8: Research Agent — Live Web Crawl + Alternate Data

8.1 Tavily Integration

(Unchanged — two-phase crawl: background deep + live refresh)

8.2 Expanded Data Sources (NEW)

MCA Portal Deep Dive:

- Annual filing compliance check (has company filed annual returns on time?)
- Director disqualification status (DIN deactivated?)
- Related party network mapping (directors common across companies)

e-Courts Comprehensive Search:

- Civil cases, cheque bounce cases, consumer disputes, arbitration
- Broader than just DRT/NCLT
- Promoter personal litigation (not just company)

Social Media & Forums:

- Twitter/X: company mentions, promoter mentions, sector discussions
- LinkedIn: company updates, employee sentiment
- Industry forums (e.g., Team-BHP for auto, MoneyControl for finance)
- Sentiment polarity scoring (positive/negative/neutral)

RBI/SEBI/Government Portals:

- Recent circulars affecting sector
- Regulatory enforcement actions (penalties, restrictions)
- Sector-wise credit growth data

Conflict Resolution: (Unchanged — surface contradictions, require manual resolution)

8.3 Research Agent Log

(Unchanged — complete transparency of queries, sources, FinBERT scores, which C each feeds)

Section 9: Frontend Architecture (NEW)

9.1 Hosted Web Application Requirements

Deployment:

- Cloud-hosted (AWS/Azure/GCP), publicly accessible URL
- No VPN required
- SSL/TLS enabled (HTTPS)
- Supports concurrent users (load balancing if needed)

Tech Stack:

- Backend: Python FastAPI (async, handles ML libraries well)
- Frontend: React + TailwindCSS (or single-file HTML if simpler)
- Database: PostgreSQL + Databricks Delta tables
- File Storage: S3 or equivalent for uploaded documents

9.2 User Journey (8 Screens)

Screen 0 — Entity Onboarding (NEW)

Multi-step form:

- Step 1: Entity details (CIN, PAN, Company Name, Sector, State, Turnover)
- Step 2: Loan details (Type, Amount, Tenure, Interest Rate, Purpose)
- Step 3: Document upload (5 core types + supporting docs)
- Step 4: Processing overlay → Dashboard

Screen 1 — Case Dashboard

- Five Cs summary strip
- Key metrics with peer benchmarks
- Alert panel (recalibration warnings, conflicts)
- Quick navigation to other screens

Screen 2 — Document Ingestor (EXPANDED)

- Document inventory table with **classification column**:
    - Auto-classified type
    - Confidence score
    - **Approve / Deny / Edit controls** (human-in-the-loop)
- **Schema Editor Panel (NEW):**
    - Left: Field configuration (add/rename/redefine fields)
    - Right: Extracted data table (editable cells)
    - Mapping interface (drag columns to schema fields)
- GST-Bank reconciliation panel (unchanged)
- Data quality summary

Screen 3 — Research Agent (EXPANDED)

- News terminal (unchanged)
- **MCA Panel (NEW):**
    - Director list with other directorships
    - Compliance status
    - Charges registered
- **e-Courts Panel (NEW):**
    - Active cases table (case type, court, amount, status)
    - Litigation density score
- **Sentiment Analysis Panel (NEW):**
    - Social media mentions chart
    - Sentiment polarity breakdown
- **SWOT Preview (NEW):**
    - AI-generated SWOT (editable before CAM finalization)
- Live refresh button
- Sources panel with URLs

Screen 4 — Five Cs Scorecard

- Composite score with weighted breakdown
- Per-C expandable accordions
- Peer benchmarking (Isolation Forest)
- Weight configuration access
- "What would change this decision" panel

Screen 5 — Graph Visualization

- Interactive network graph (Cytoscape.js)
- Node detail panel
- Graph risk summary (3-5 flags)

Screen 6 — Primary Notes Portal

- Structured input form per C
- Live score impact display
- Override audit trail

Screen 7 — CAM Preview and Export (EXPANDED)

- Full CAM rendered (10 sections now, includes SWOT + Pre-Cognitive)
- **SWOT Matrix (NEW):**
  - 2×2 grid (Strengths, Weaknesses, Opportunities, Threats)
  - **Editable** — user can click any quadrant to add/edit/delete points
  - AI-generated content shown with "AI-suggested" tag, user edits shown as "User-added"
- Hover-to-source (unchanged)
- Export to Word/PDF (now includes SWOT matrix)
- Audit trail table

9.3 Design Language

PRESERVED from original design:

- Dark theme (#0a0a0a background)
- Amber (#f59e0b) and blue (#3b82f6) accent colors
- JetBrains Mono (code/data) + Syne (headings) fonts
- Cursor ring effect
- Scanline + noise texture
- Terminal-style components

Section 10: Evaluation Criteria Mapping

Criterion	How Intelli-Credit Addresses
Operational Excellence	Hosted web app, stable upload, processing animations, error handling
Extraction Accuracy	AWS Textract + human validation + schema editor with editable cells
Analytical Depth	FinBERT + Isolation Forest + Graph + Pre-Cognitive + Tavily + MCA + e-Courts + sentiment
User Experience	Multi-step onboarding, approve/deny/edit classification, live SWOT editing, hover evidence

Explainability	Every signal traceable to source, audit trail, "what would change" output, hover tooltips
Indian Context	GST-bank reconciliation, promoter pledge %, DRT cases, MCA charges, ALM gaps, CIBIL data
Secondary Research	Tavily news + MCA + e-Courts + social sentiment + macro indicators
SWOT & GenAI	LLM-generated editable SWOT, comprehensive downloadable CAM report

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Section 11: Innovations Beyond Problem Statement

(Expanded from original 7 innovations)

Innovation 1 — Isolation Forest Peer Benchmarking (unchanged — trained on Screener.in data)

Innovation 2 — Graph Relationship Engine (unchanged — director networks, group concentration)

Innovation 3 — Three-Level Weight Configurability (unchanged — policy + signal + per-case)

Innovation 4 — Recalibration Warning System (unchanged — second approver for overrides that flip decision)

Innovation 5 — "What Would Change This Decision" (unchanged — constructive path to approval)

Innovation 6 — Confidence Scoring on Extracted Values (unchanged — <75% flagged)

Innovation 7 — Research Conflict Resolution (unchanged — contradictions surfaced, not silently overwritten)

Innovation 8 — Human-in-the-Loop Classification (NEW)

- AI classifies, human approves/denies/edits before analysis
- Prevents garbage-in-garbage-out from misclassified documents

Innovation 9 — Dynamic Schema Configuration (NEW)

- User-configurable extraction schema
- Handles non-standard document formats
- Cell-level editing for OCR corrections

Innovation 10 — Pre-Cognitive Analysis (NEW)

- Velocity trends, triangulation flags, sentiment divergence
- Forward-looking risk signals beyond traditional metrics

Innovation 11 — Editable SWOT Matrix (NEW)

- AI suggests, user refines
- Collaborative intelligence approach

Innovation 12 — 360-Degree Alternate Data (NEW)

- Beyond news: social sentiment, macro indicators, MCA deep dive, comprehensive e-Courts
- Most comprehensive secondary research in the space

Section 12: Technology Stack Summary

Component	Technology	Why
OCR and document parsing	AWS Textract	Best for messy Indian PDFs, handles tables
Data orchestration	Databricks + Delta tables	Required by PS, parallel jobs, lineage
NLP risk extraction	FinBERT (HuggingFace)	Pre-trained on financial text, zero-shot
Anomaly detection	Isolation Forest (scikit-learn)	Unsupervised, interpretable, fast
Graph analysis	NetworkX (Python)	Lightweight, handles MCA data well
Graph visualization	Cytoscape.js	Interactive, web-native
Web research	Tavily API	Purpose-built for AI agents, returns content
Sentiment analysis	HuggingFace Transformers (DistilBERT)	Pre-trained sentiment classifier
LLM narrative	Claude API / GPT-4	Grounded generation from structured inputs
CAM export	python-docx	Programmatic Word doc generation
Backend	Python (FastAPI)	Async, ML library integration

Frontend	React + TailwindCSS	Component-based, responsive
Database	PostgreSQL + Databricks Delta	ACID transactions, scalability
Deployment	AWS/Azure/GCP	Cloud-hosted, public URL
Training data	Screener.in	Indian company financial data

Section 13: Gaps Explicitly Closed

From Original PS:

- ✅ Board Meeting Minutes (FinBERT governance extraction)
- ✅ Shareholding Pattern (promoter pledge % flagging)
- ✅ Sanction Letters (existing facilities, encumbrances)
- ✅ Live Web Crawl (Tavily two-phase architecture)

From Updated PS:

- ✅ Entity Onboarding (CIN, PAN, loan details capture)
- ✅ 5 Core Document Types (ALM, Shareholding, Borrowing Profile, Annual, Portfolio Cuts)
- ✅ Human-in-the-Loop (approve/deny/edit classification)
- ✅ Dynamic Schema (configurable extraction mapping)
- ✅ SWOT Analysis (AI-generated, user-editable)
- ✅ Alternate Data (sentiment, macro, comprehensive e-Courts/MCA)
- ✅ Hosted Web App (cloud deployment, public URL)

END OF UNIFIED REQUIREMENTS SPECIFICATION

This specification now fully integrates both problem statements while maintaining the depth, structure, and explainability of the original. Every new requirement from the updated PS is architecturally mapped to the existing four-layer system, with clear data structures, processing logic, and UI components defined.